LUMEIA

EXPERIENCE SPACE. DISCOVER YOU

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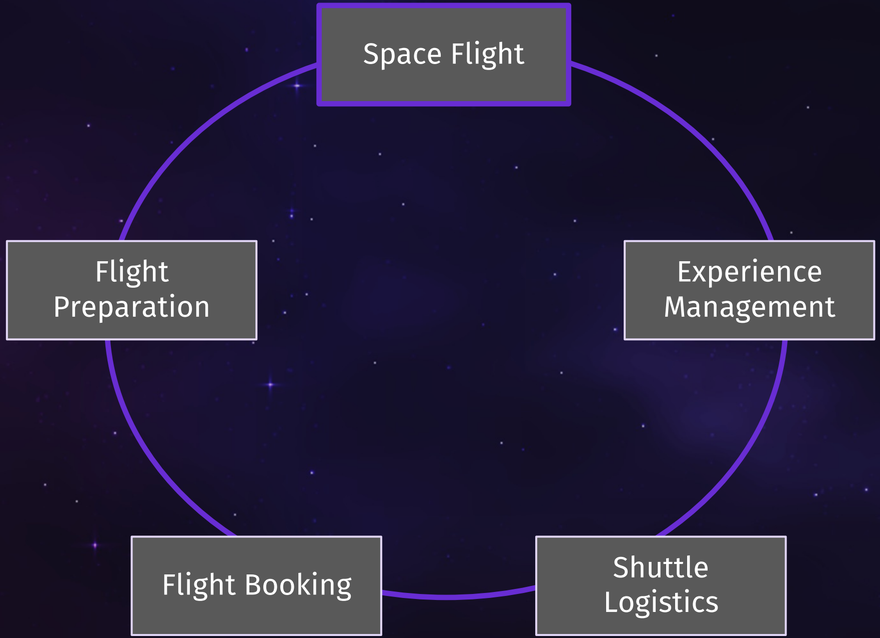
Introduction

„I need your help to fulfil my vision of space tourism !“

Elon Bezos

The company is fighting bankruptcy and needs fast, reliable support

That's where we IBAITs come into play



Agenda

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Our Task

02
Storyline

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01

Our Task

Issues & Goal

Problems

- High mental pressure
- Worthiness
- Uncertainty
- Fear and worry
- High physical strain

Goal

- Customer satisfaction
  - Reduce stress
  - Increase safety
    - Open communication
    - Secure software
    - Medical support

02

Storyline

Herangehensweise

Overview
Goal

Problems
Requirement

Approach
Possible solutions

Implementation
Final solution

Personas

Ben Cooper (Passenger)

- Carpenter
- Saved up all his money for the flight

Motivation:

- Wants to fulfil his lifelong dream

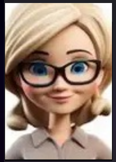
- Limited Funds
- Scared of not fitting in the group

Fear and worry

High mental pressure

Reduces stress

Open communication



Gemma Palmer (Crew)

- Working as a psychologist at the Base station
- Supports everyone in the rocket with medical concerns

Motivation:

- Bring customers to and from space safely

Personas

Peter Meyer (Passenger)

- Kid of a wealthy man/woman
- Was dragged into all this

Motivation:
- Please parents
- Brag at school

Barries & Frustrations:
- Didn't want to go
- Doesn't understand whats so great about it

Anne Bright (Crew)

- Stewardess of the rocket

Motivation:
- Bring customers to and from space safely
- Bring customer satisfaction

Barries & Frustrations:
- High responsibility
- Turbulences and unpredicables

Remote Personas

Brendon Fitz

- Working as Crew of the Base Station

Motivation:

- Bring customers to and from space safely

Barriers & Frustration

- Turbulences and unpredictables
- High responsibility for rocket

Dave Swington

- Doctor at the Base Station
- Supports everyone in the rocket with medical concerns

Motivation:

- Bring customers to and from space safely
- Support medical indidents

Barriers & Frustration

- Not directly in the rocket (remote)

Experience Modes

Relax

- Calm visual styling
- Focus on safety
- Fewer stimuli
- Pshychological help button

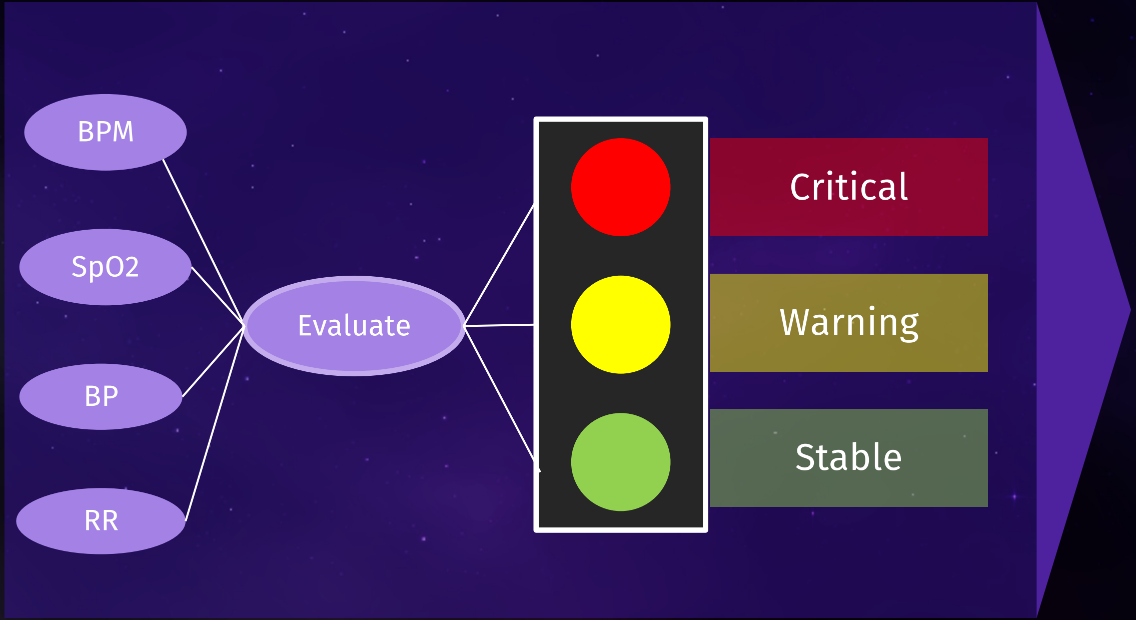
Normal

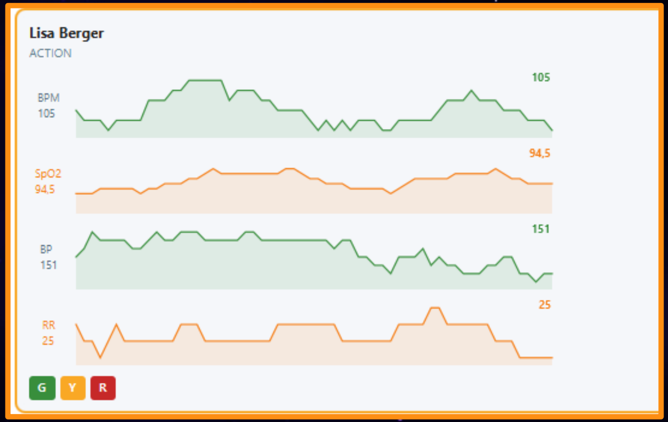
- Casual visual styling
- Clear orientation
- No special options

Action

- Darker visual styling
- Adrenaline experience
- Experience factor and memory factor

AI Health


BPM
SpO2
BP
RR
Evaluate
Critical
Warning
Stable


Lisa Berger
ACTION
105
BPM
105
94,5
SpO2
94,5
151
BP
151
25
RR
25
G
Y
R

AI Health Functionality

Training_data_csv: 140 different example-patients labled:
GREEN, YELLOW, RED

KNN

- Calculate similarity from vital stats and training data (eucledean distance)
- Demographic-Bonus when test case has same AgeGroup, Gender or Experience Mode
- Take k=5 and decide the result

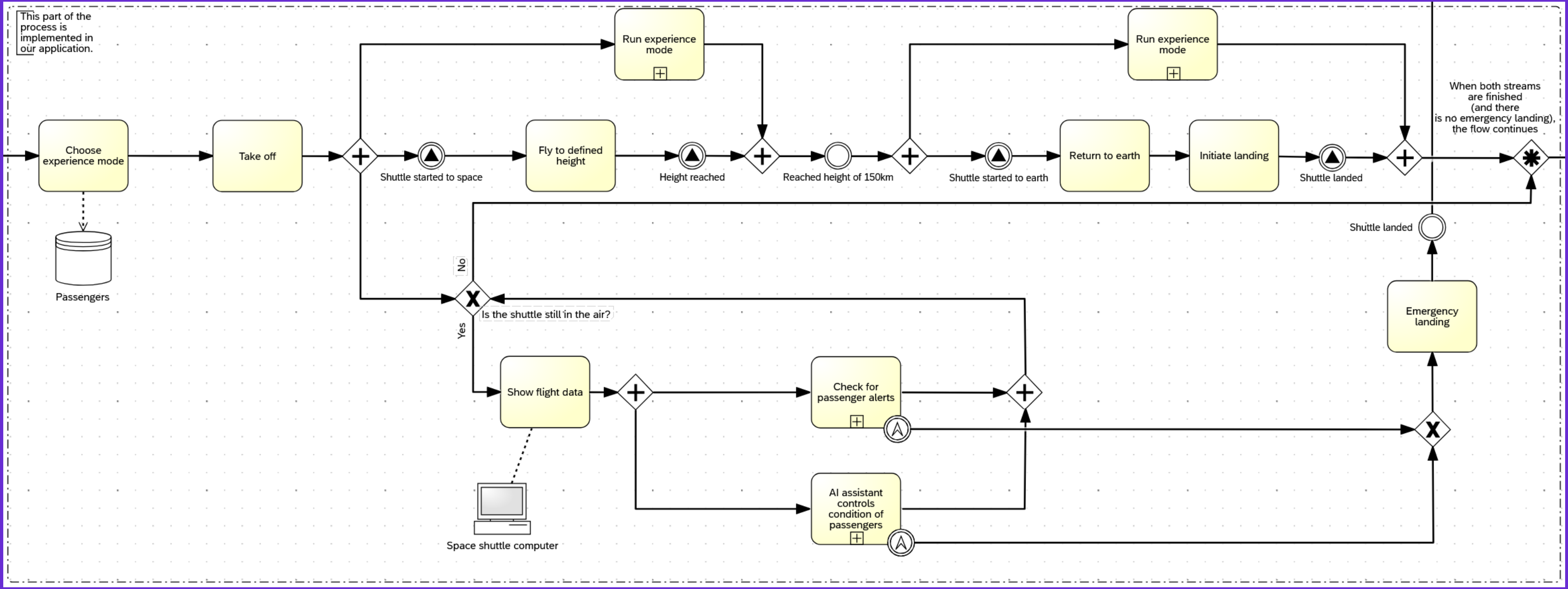
Recommendation for overall status

Z-Score

- Checks every single vital stat on its own
- Vital Profile Table includes 18 Demography Groups with mean and stdDev
- How many stdDev is the stat away from mean
- Thrsholds change in ascent and descent

Escalates results from KNN

BPMN (Business Process Model and Notation)



03

Value

What makes us different?

You don't have to be an astronaut to experience space

Return happy, safe & transformed

- Always understood
- Always watched over
- Always personal

Our unique selling propositions

Experience Modes

Health Monitoring

User Interfaces

Psychology

Our added value

Drastically Increased Security

Trust & Comfort for Passengers

Efficiency & Automation

Scientific Progress

Setting the Standard for the Future

04

Live-Demo

Live Demo – Process and Data Exchange

In this demo, we show:

- How flight and vital data are updated continuously
- How passenger actions trigger system processes
- How information is transferred between dashboards
- How incidents are escalated from passenger to crew

Layered Architecture

1

Bootstrap / Composition Root

app (Launcher, SpaceFlightApp, AppContext)

2

Presentation

ui. (passenger, basestation, aihealth, shared)*

3

Service / Use-Case

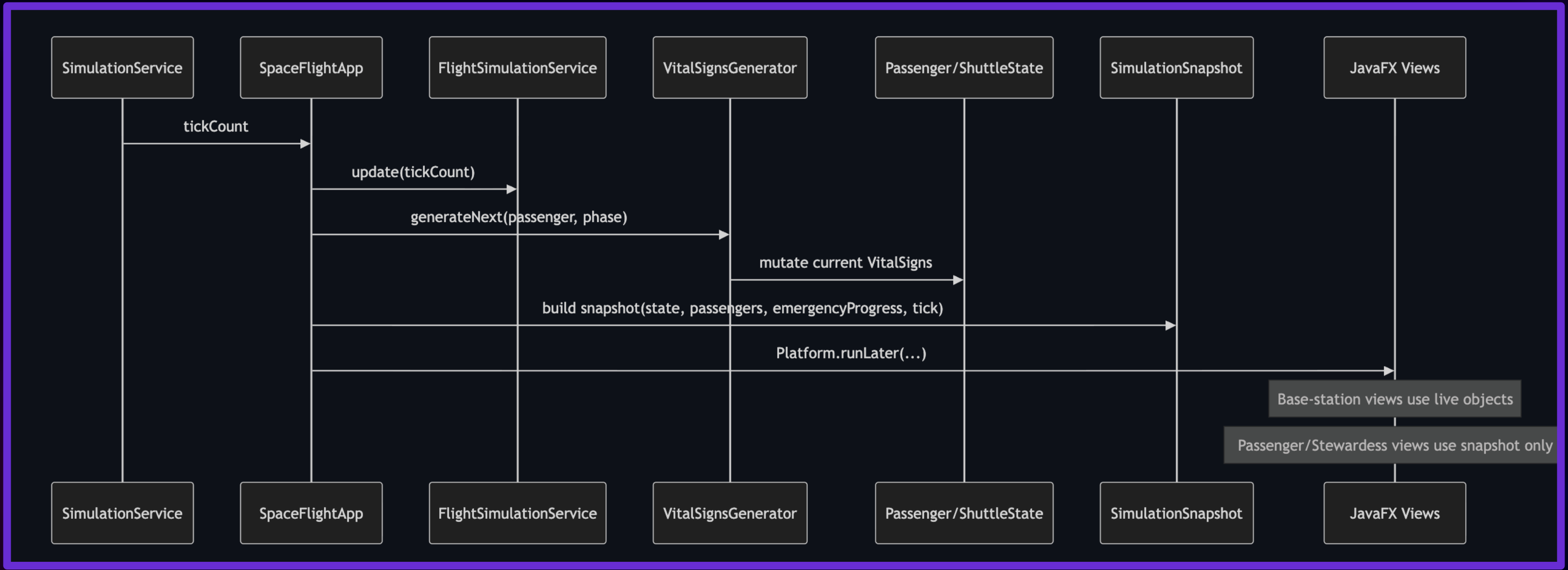
simulation · health · alert

4

Domain Model

model (Passenger, ShuttleState, enums, VOs)

Runtime Dataflow



05

Lessons Learned

Lessons Learned from the project

1. Technical and Tool Skills

Github: version control andcollaboration

IntelliJ: main development environment

Java, JavaFX, and Maven: project setting

Lessons Learned from the project

2. Software Engineering Practice

- Applying clean structure and common practices
- Better understanding of SOLID and maintainable code
- Working with UI design, mockups, technical documentation, user documentation, GitHub wiki pages and so on.
- Seeing fast the full software development lifecycle, not just coding

Lessons Learned from the project

3. AI-Assisted Development

- AI and agent AI
- Human responsibility
- Different tools
- Generate, explain & work

Lessons Learned from the project

4. Teamwork and Project Organization

Coordinating tasks and responsibilities

Agile Work

Time pressure

Use strengths of each team member effectively

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